Earth and Planetary Science Letters, 35 (1977) 92–104
© Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

AGE OF GRENVILLE BELT MAGNETISATION: Rb–Sr AND PALAEOMAGNETIC EVIDENCE FROM SWEDISH DOLERITES

P.J. PATCHETT [1]

Scottish Universities Research and Reactor Centre, East Kilbride, Glasgow G75 OQU (Scotland)

and

G. BYLUND

Department of Mineralogy and Petrology, University of Lund, Sölvegatan 13, S-223 62 Lund (Sweden)

Received November 18, 1976
Revised version received February 28, 1977

Rb–Sr mineral ages and palaeomagnetic data are presented for nine dolerite dykes and sills in central and southern Sweden. The intrusions are representative of widespread dolerites, mainly dykes, which cut pre-Sveconorwegian (Svecofennian) crust and trend parallel to the Sveconorwegian Front Zone (or Schistosity Zone) over a band up to 150 km wide along most of its 700-km length. The Rb–Sr ages suggest emplacement through the approximate interval 1000–900 m.y. Cooling ages on country rocks show little evidence of significant Sveconorwegian (e.g. 1000 m.y.) heating effects, and the new isotopic and palaeomagnetic results are taken to represent the time of dolerite intrusion.

The dolerites were emplaced in response to net tensional stresses in the Svecofennian crust: it is proposed that these were caused by marginal upwarping related to post-orogenic uplift of the adjacent Sveconorwegian region.

The palaeomagnetic poles for the dolerites are similar to poles from the Sveconorwegian Province and the Front Zone, indicating that these also have a ca. 1000–900-m.y. age. The similar cooling histories of the Sveconorwegian Province and the Grenville Province of Canada allow this age to be assigned also to the Grenville poles. If so, it is not necessary to postulate plate motions during Grenville-Sveconorwegian magnetisation, and the agreement between dolerite poles and Sveconorwegian poles suggests that the orogenic belt cooled essentially in place against its bounding cratons. A possible continental fit shows Scandinavia and North America contiguous in late Grenville time, though relative motions before 1000 m.y. are not ruled out.

1. Introduction

The Grenville Province of Canada comprises rocks that yield K–Ar ages on whole rocks and minerals in the range 1150–800 m.y., though the large majority of determinations fall in the 1050–850-m.y. interval [1,2]. The Grenville Province is shown structurally [3,4] and geochronologically [5,6] to be composed in large part of Archean or Lower Proterozoic rocks, but the K–Ar ages remain a very clear unifying feature for the whole belt.

The Grenville has an equivalent in the Sveconorwegian Province of the Baltic Shield [7]. Here a variety of rocks, at least partly of much older age [8,9], all exhibit K–Ar ages between ca. 1100 and 800 m.y., though as for the Grenville, the main concentration is between 1050 and 850 m.y. [10–12]. The K–Ar ages, widely interpreted in orogenic belts as documenting uplift and cooling [1,7], demonstrate that the Grenville and Sveconorwegian Provinces are parts of a single coherent belt which was uplifted and cooled from its latest thermal event during the general interval 1050–850 m.y., though the precise limits are uncertain. The Grenville and Sveconorwegian belts are separated from older regions by rather distinct, apparently subvertical tectonic boundaries, in Canada the Grenville Front (Fig. 5), in Scandinavia the Sveco-

[1] Present address: Institut für Mineralogie, Universität Münster, Gievenbecker Weg 61, D-4400 Münster, West Germany.

norwegian Front or Schistosity Zone (Figs. 1, 5). K–Ar ages from rocks lying beyond these boundaries are much older than 1000 m.y., and show that except for a narrow band adjacent to the "fronts", Grenville-Sveconorwegian thermal effects did not extend significantly into the pre-Grenville/Sveconorwegian crust. The ultimate age and origin of these tectonic boundaries remains a matter for debate [5,8,13,14].

Palaeomagnetic pole positions from Canada for rocks thermally affected during the Grenville, published by many authors, define a distinct belt of apparent polar wander, the "Grenville track" (see Fig. 4) which does not correspond to any other poles from North America. Poles from the pre-Grenville regions are reasonably abundant down to ca. 1100 m.y., the probable age of the younger Keweenawan results (see Irving and McGlynn [15] for a comprehensive review). The widespread Franklin basic igneous events yield well-controlled data at ca. 675 m.y. [16], but very few reliably dated poles are known for the period between 1100 and 675 m.y. Almost no data exist for the age range 1000–800 m.y., so that the polar wander path of pre-Grenville North America during this period is very uncertain [15]. Since the poles for the Grenville Province are derived from rocks with a protracted metamorphic and cooling history, it has not been possible to define their ages even to within 200 m.y., and the various models proposed to explain the lack

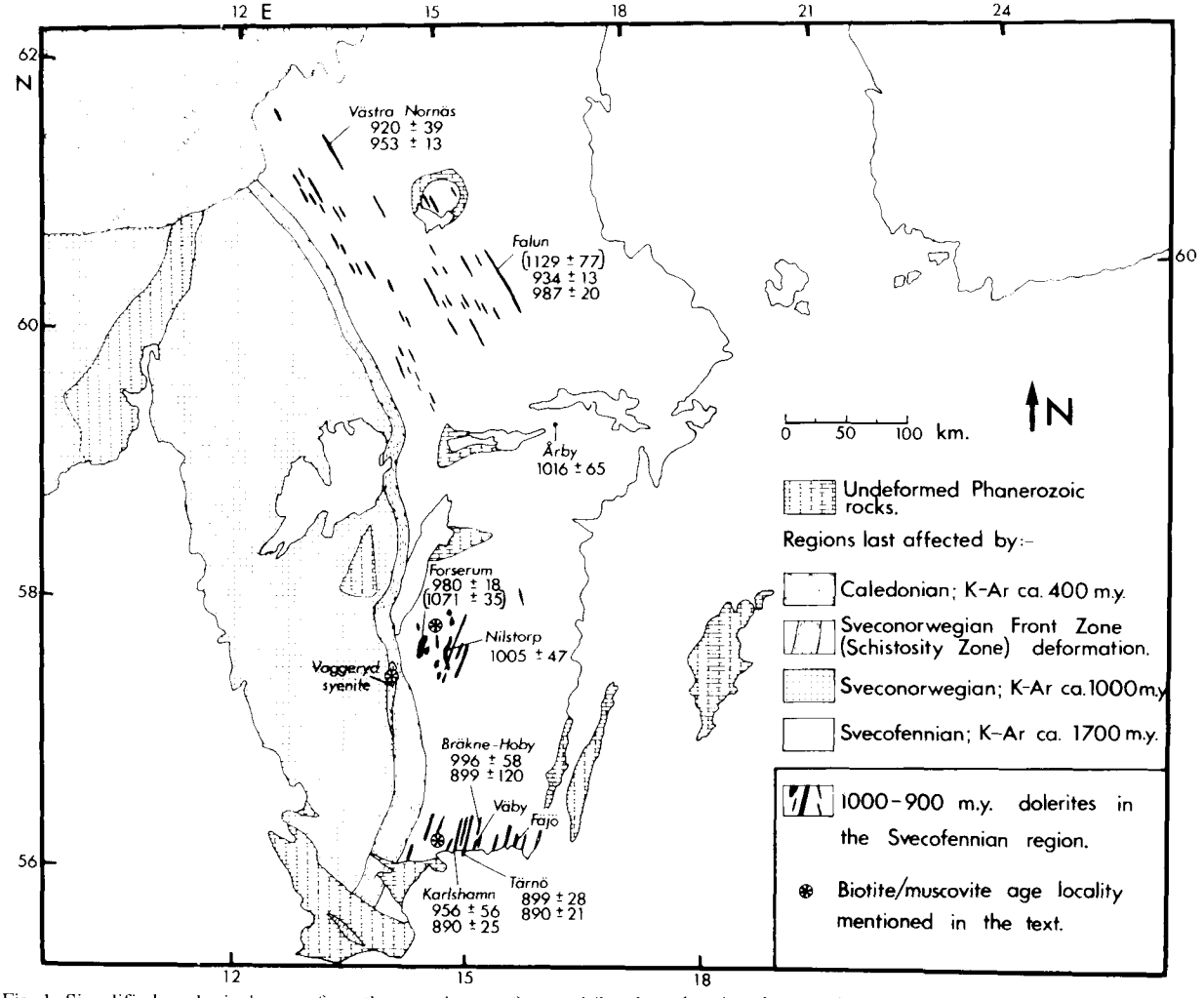


Fig. 1. Simplified geological map of southern and part of central Sweden, showing the sampled localities with Rb–Sr mineral ages.

of correspondence between poles from the Grenville and pre-Grenville regions have been summarised by Irving et al. [17] and Ueno et al. [18].

One model assumes a rather old age for some or all Grenville poles, corresponding in a general way to Rb–Sr whole rock ages of 1200 m.y. or more [19–21]. This age assignment results in the Grenville poles defining an additional pre-1200-m.y. loop in the polar wander path, unrepresented in any rocks from the pre-Grenville region.

A second model assumes an 1120–1000-m.y. age for the Grenville track, and hence essential contemporaneity with the mid-Upper Keweenawan track from the pre-Grenville region. Since the two polar paths do not correspond, but appear to meet at an approximate Keweenawan age of 1000 m.y., this age assignment implies a convergence between a separate continent "Grenvillia" and the pre-Grenville shield during 1120–1000 m.y., with collision occurring at ca. 1000 m.y. along a hypothetical suture within the Grenville Province [22,23,17,15]. Proof of this model would demonstrate the operation of present-day style orogenic processes in the Grenville belt.

A third model assumes that the Grenville poles are of ca. 1000–800-m.y. age, in general agreement with K–Ar ages from the belt. This would imply that the Grenville poles define a post-Keweenawan loop that is not represented in data from elsewhere in North America [24], a hypothesis consistent with the lack of poles of 1000–800 m.y. age from the pre-Grenville region. This model receives support from new data showing that the magnetisation for poles on the "Grenville track" was probably acquired *after* Grenville-age folding [25]. Additional circumstantial support is provided by the fact that on a hypothetical continental fit, the Grenville poles correspond to dated poles of 1000–800 m.y. from Africa [26,27].

The problem of the history of the Grenville-Sveconorwegian Province, as it concerns palaeomagnetism and related geochronology, is therefore made up of two factors: (1) the ages of the Grenville and Sveconorwegian poles are not independently known, and (2) there is not a complete palaeomagnetic record for the pre-Grenville/Sveconorwegian regions from 1400 to 700 m.y., hence the ages of the Grenville and Sveconorwegian poles cannot be approximated by comparison with dated poles.

This study discusses Rb–Sr mineral age data and palaeomagnetic poles for dolerites, chiefly dykes which intrude the pre-Sveconorwegian (Svecofennian) region of Sweden; the width of the dykes (or thickness in the case of sills) ranges from 5 to 250 m, and they consist of a variety of olivine- and/or hypersthene-bearing dolerites [28]. The dykes trend consistently parallel to the Sveconorwegian Front or Schistosity Zone which separates the Sveconorwegian Province from the Svecofennian, and occur along almost its whole 700-km length over a belt up to 150 km in width from the Front eastwards (Fig. 1). The dykes are vertical or subvertical, and show no evidence of post-intrusive tilting.

2. Ages of dolerite intrusion

Samples from eight intrusions, six dykes and two sills, were separated into a series of feldspar density fractions (up to nine) plus clinopyroxene and/or apatite. In one intrusion where felsic xenoliths recrystallised by the dolerite were used, epidote and sphene replace clinopyroxene and apatite. General analytical techniques are described elsewhere [29]: most samples from dolerites in this study have an internal 1σ error on $^{87}Sr/^{86}Sr$ of between 0.00003 and 0.00010, and for regression analysis the average of these "in-run" errors for all the points on the line together with an estimated 1σ uncertainty of 0.7% on $^{87}Rb/^{86}Sr$ (based on previous replicate analysis) is used. Biotites and samples from felsic xenoliths differ in that an estimated uncertainty of 0.04% is used for $^{87}Sr/^{86}Sr$. Mineral methods are used in preference to whole rock methods because whole rocks have a low range of Rb/Sr ratios and because in many cases upward bias of whole rock lines is observed due to crustal contamination of the magmas [30].

This paper is intended only to summarise the age results for the dykes parallel to the Sveconorwegian Front; full data for all post-Svecofennian dolerites studied, with isochron diagrams, will be given elsewhere [31]. The age results are listed in Table 1 and illustrated in Fig. 1. The decay constant 1.39×10^{-11} yr^{-1} is used and errors are given at the 2σ (95% confidence) level.

The value of the mean of the squared weighted deviates (MSWD), expressing degree of scatter of data

TABLE 1

Summary of Rb–Sr mineral age data for dolerites

Intrusion	Sample	Method	Number of points	Age (m.y.)	MSWD
Karlshamn	BL-3	dolerite minerals	8	956 ± 56	2.07
Karlshamn	BL-18	dolerite minerals	7	890 ± 25	3.72
Tärnö	BL-2	xenolith minerals	5	899 ± 28	2.22
Tärnö	BL-ARX	xenolith minerals	4	890 ± 21	2.22
Bräkne-Hoby	BL-24	dolerite minerals	6	996 ± 58	1.85
Bräkne-Hoby	BL-27	dolerite minerals	6	899 ± 120	4.41
Forserum	AL-6	dolerite minerals	7	980 ± 18	2.89
Forserum	AL-12	dolerite minerals	7	1071 ± 35 *	4.71
Nilstorp	AL-22	dolerite minerals	7	1005 ± 47	7.56
Årby	SÖ-50	dolerite minerals	7	1016 ± 65	5.43
Falun	DS-9	dolerite minerals	6	987 ± 20	1.06
Falun	DS-31	dolerite minerals	6	1129 ± 77 *	7.34
Falun	DS-31	whole rock biotite	2	934 ± 13	
V. Nornäs	D75-26	dolerite minerals	7	920 ± 39	6.25
V. Nornäs	D75-26	whole rock biotite	2	953 ± 13	

Number of points = number of Rb–Sr data points defining the best-fit regression line.
* Ages considered to be anomalously old; see text.

points about the best-fit line, is given for each regression age (Table 1). Some MSWD values are below the postulated upper limiting value of ca. 2.5 for an isochron relationship between data points [32], but several others lie above this figure. This could be due to real geological scatter of the points in excess of analytical uncertainty, but may be partly or wholly caused by the method of assigning the means of in-run $^{87}Sr/^{86}Sr$ errors to the lines as estimated analytical uncertainties: it is possible that this error estimate, while being the only systematically calculated uncertainty that could be used in the present study, is actually too low, thus yielding artificially high MSWD values. None of the dolerite ages discussed in this paper are considered to be unreliable on statistical grounds.

Except for one apatite from Forserum, clinopyroxene and/or apatite are included in the regression treatments and are concordant with the feldspars. The results demonstrate essential agreement of dolerite mineral lines, xenolith mineral lines and whole rock–biotite pairs with a general 1000–900-m.y. age for the dolerites, except in two cases. The dolerites of Forserum and Falun both exhibit mineral lines over 1050 m.y. which fail to agree at 2σ with other, younger results from the same intrusions. This is considered to be due to contamination of late-crystallising alkali feldspar by crustal, radiogenically enriched Sr, leading to upwardly biased mineral lines [30], and the two older results are therefore discarded as spurious.

Since this contamination effect could have operated to a lesser extent for other dolerite mineral lines, small differences in age from one intrusion to another cannot be interpreted literally. If the simplest interpretation were followed, all age results would be regarded as comparable. The spread of results between 1020 and 890 m.y. is, however, believed to reflect at least in part a real variation in intrusive age: in particular, ages of less than 900 m.y. are recorded only from the Karlshamn and Tärnö dykes, and it is *possible* that these are younger than other members of the "swarm".

The country rocks of the dykes are usually of Svecofennian or late Svecofennian age [33–35], but the sills of Forserum and Nilstorp, as well as the dyke of Västra Nornäs penetrate unmetamorphosed post-Svecofennian continental sediments. In central Sweden several K–Ar ages show that the country rock has not been thermally disturbed since ca. 1700 m.y. [10]. Rb–Sr ages on biotite and K-feldspar [31] from country-rock granite 20 km east of the Sveconorwegian

Front Zone near Forserum (Fig. 1) give ca. 1650 m.y., a reasonable late-Svecofennian cooling age. Biotite from granite mid-way between the Karlshamn dyke and the Front Zone (Fig. 1) gives ca. 1380 m.y., an age consistent with the age of intrusion of this (Karlshamn) group of granites at 1455 m.y. [36]. These Rb–Sr mineral and K–Ar ages show that Sveconorwegian (1100–900 m.y.) thermal effects are essentially absent from all the regions of study; the mineral results from the dolerites therefore probably reflect intrusive ages between 1000 and 900 m.y., and not some isotopic resetting event.

3. Palaeomagnetic study

Oriented hand-samples and drill-cores were collected from the same intrusions as were covered by the Rb–Sr age work. In addition palaeomagnetic samples were obtained from dolerites at Fäjö and Väby (Fig. 1), while Forserum was omitted from the palaeomagnetic study due to the extensive faulting in this intrusion. Sample orientation was performed with a magnetic compass in conjunction with topographical sighting. In addition, to check for possible magnetic influences from the outcrop, comparison was made between readings obtained with the compass at different levels of the exposure and sighting along some prominent feature. No samples were taken in the few cases where differences between readings exceeded 5°.

One to three cylindrical specimens 2.50 cm in diameter and 2.25 cm long were obtained from each sample and measured on a high-speed spinner magnetometer, the Czech JGF-3. Sample means were used for calculating the mean magnetic directions for the different intrusions. To test the magnetic stability of the specimens the usual alternating field (AF) and thermal demagnetisation techniques were applied. At least two specimens from each site were demagnetised in 13 steps to a peak field of 1000 Oe. During all AF demagnetisation the specimens were rotated in a two-axis tumbler. It was found that peak alternating fields of 200–400 Oe were sufficient to produce a stable end-point of the magnetic vector, and the remaining specimens were therefore demagnetised in a few steps to a peak field of 500 Oe. At least one specimen from each site was thermally demagnetised in steps of 50°C to 550–650°C.

The results are shown in Table 2, and Fig. 2 gives examples of the behaviour during demagnetisation of pilot specimens from the intrusions of Karlshamn and Tärnö, plotted both as directions and in terms of magnetic intensity (J) relative to intensity before demagnetisation (J_0). The demagnetisation patterns of specimens from other dolerites were very similar to those of Fig. 2, with the exception of samples from the dyke at Västra Nornäs, which presented no stable end-point during demagnetisation.

The thermal demagnetisation curves of the dolerites (Fig. 2b) with their well-marked "shoulders" give blocking temperatures of 450–550°C, indicating that magnetite or titanomagnetite is the main carrier of the magnetic remanence. This is also suggested by ore microscopy, which proved the presence of magnetite with exsolution lamellae of ilmenite together with single grains of ilmenite, magnetite and occasional sulphide.

The single blocking temperature interval of the thermal demagnetisation curves (Fig. 2b) suggests that the remanence of the dolerites was acquired as a thermal remanent magnetisation (TRM) when the rocks cooled, and has not been affected by any later thermal events [37]. Support for this is given by results from baked red sandstone xenoliths in the Tärnö intrusion. These have a slightly higher blocking temperature than the magnetite/titanomagnetite-bearing dolerites (Fig. 2b), consistent with haematite being the main carrier of their remanence [37], and the magnetic directions of dolerite and sandstone are in close agreement at all except the most extreme conditions of demagnetisation (Fig. 2a and b). Since magnetite and haematite would be expected to respond to a later resetting event in different ways [37], then this agreement suggests that both sandstone xenoliths and dolerite acquired their present remanence during post-intrusive cooling of the intrusion and have remained magnetically undisturbed since that time. Evidence that the Tärnö dolerite cooled quickly, and therefore had a magnetic blocking temperature interval comparable to that observed in the laboratory is provided by its having intruded, and become intimately intermixed with poorly consolidated, possibly water-laden sediments (A. Rodhe, personal communication, 1976). The Nilstorp dolerite also penetrates unmetamorphosed continental sediments, as is the case for the Forserum and Västra

TABLE 2

Mean magnetic directions of dolerites after demagnetisation with corresponding virtual geomagnetic pole positions

No.	Intrusion	n/N	AF (Oe)	Decl. (deg.)	Incl. (deg.)	α_{95} (deg.)	k	R	Virtual pole position		dp (deg.)	dm (deg.)
									lat.	long.		
1	Karlshamn	14/9	300	129.8	81.0	4.8	116	8.9312	43.2°N	33.5°E	8.9	9.3
2	Tärnö	14/5	300	312.8	−41.9	4.8	259	4.9846	0.3°S	57.0°E	3.6	5.9
3	Bräkne-Hoby	16/7	400	309.3	3.6	6.1	98	6.9387	22.2°S	71.9°E	3.1	6.1
4	Väby	7/3	200	123.1	27.2	(19.5) *	–		5.0°S	69.6°E	–	–
5	Fäjö	10/6	200	312.6	2.6	5.4	157	5.9682	23.3°S	68.9°E	2.7	5.4
6	Nilstorp	18/6	400	315.2	−27.4	9.7	49	5.8971	9.0°S	58.5°E	5.8	10.6
7	Årby	17/6	200	142.5	52.4	7.0	93	5.9460	7.3°N	47.4°E	6.8	9.5
8	Falun	16/10	200	131.4	45.8	5.6	75	9.8795	6.1°N	57.6°E	4.6	7.1
9	V. Nornäs	9/3	–	–	–	–	–	–	–	–	–	–

Numbers refer to poles plotted in Figs. 4 and 5. n/N = number of specimens/samples per intrusion. AF = alternating field demagnetisation level with least scatter in magnetic direction between samples. dp and dm are the semi-axes of the oval of confidence about the pole at the 95% probability level. All statistics based on sample means according to Fisher.

* Number of samples too low for statistical use.

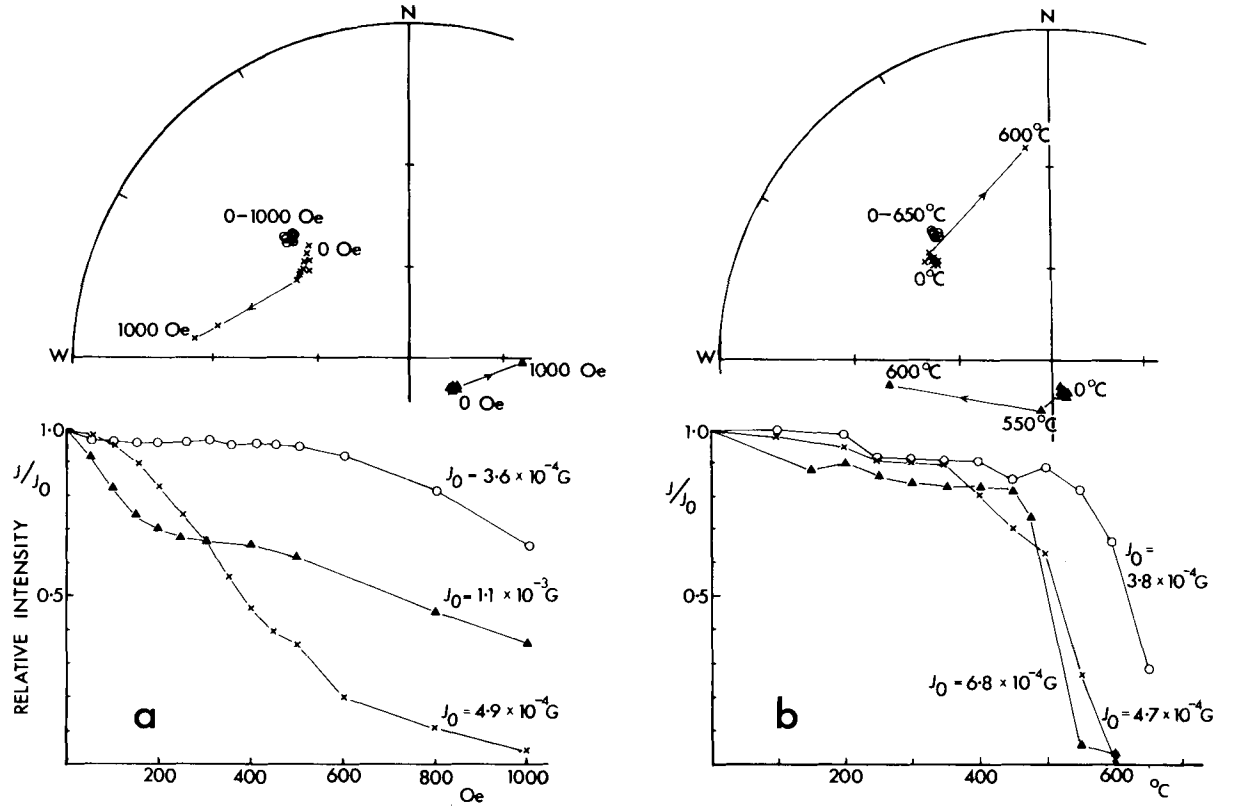


Fig. 2. Behaviour of pilot specimens from the Karlshamn and Tärnö dykes during (a) alternating field and (b) thermal demagnetisation. ▲ = Karlshamn dolerite, × = Tärnö dolerite, ○ = Tärnö sandstone xenolith. Magnetic directions are plotted on a Wulf stereographic net.

Nornäs intrusions for which no pole positions are presented. There is no evidence to suggest a protracted cooling history for any of the dolerites of this study.

4. Dykes and post-orogenic uplift

Except in one case where flat-lying continental sediments are intruded by sills, the dolerites mainly occur as dykes intruding basement and lying parallel to the Sveconorwegian Front Zone. They occur in a well-defined band over a distance of up to 150 km from the Front Zone eastwards. Fractures running parallel to the dykes are reported from several areas (e.g. [38,39]).

The consistency of strike strongly suggests that the emplacement of the dykes is related to a widespread stress pattern in the Svecofennian crust during 1000–900 m.y., and the parallelism with the Sveconorwegian Front Zone additionally implies that the dykes, the stress system and the Front Zone are all interrelated. The simple trend system of the dykes and related fractures seems to argue in a straightforward way for net tensional stress in the Svecofennian region at right angles to the Front Zone. An explanation for this is suggested by the K–Ar ages from the Sveconorwegian block, mainly falling in the range 1050–850 m.y. These are due to cooling of the metamorphic and plutonic rocks through argon blocking temperatures, a process generally associated, at least in its later stages, with uplift and erosional unroofing [1,7]. It can therefore be inferred that the Sveconorwegian block was being uplifted relative to the essentially static, thermally unaffected Svecofennian block during at least the interval 1000–900 m.y. The uplift may have begun earlier, and certainly continued later, as given by K–Ar ages down to 800 m.y. [12]. Since the Front Zone separates the two blocks between

which a break in K–Ar age occurs (see [10]), then a large part of the relative movement must have taken place across its 10–25-km width, and this could be consistent with the vertical shearing deformation which it displays. That the Front Zone was last cooled in late Sveconorwegian time is shown by mica ages from sheared rocks of the Vaggeryd syenite (Fig. 1), giving 877, 894 and 923 m.y. for biotites and 934 and 948 m.y. for muscovites (all ± 13 m.y.; for data see [31]).

The Front Zone cannot have been able to take up all the movement involved, however, and stress systems are likely to have been set up in the neighbouring Svecofennian region; the probable effect would be a partial uplift of the Svecofennian crust adjacent to the Front Zone. This would result in at least the deeper levels of the Svecofennian crust, by virtue of their upwarping and consequent stretching, being subjected to tensional stresses oriented at right angles to the Sveconorwegian margin: beyond a certain distance, presumably related to the thickness and rigidity of the Svecofennian crust, this effect would die away.

As the dykes lie parallel to the Front Zone, suggesting such a tensional stress regime, and occur over a distinct band adjacent to the Front, it seems reasonable to relate their emplacement to this mechanism, of which Fig. 3 is an exaggerated representation. The model proposes a genetic relationship between uplift and cooling of the Sveconorwegian belt on the one hand, and fracturing and dyke emplacement in the Svecofennian region on the other. The warping of the Svecofennian crust cannot have been great, because it

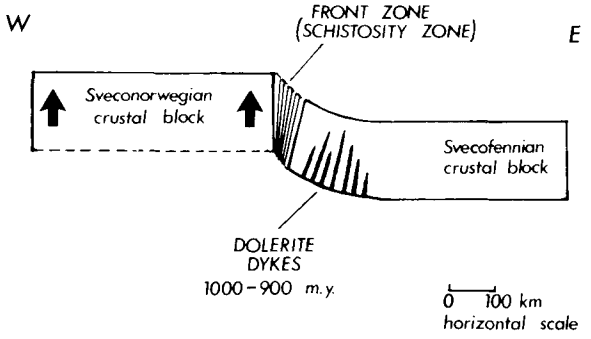


Fig. 3. Exaggeration of the postulated tectonic scheme relating post-orogenic uplift of the Sveconorwegian belt to fracture and dolerite emplacement in the Svecofennian crust during 1000–900 m.y.

apparently did not lead to significant tilting of early emplaced dykes and consequent departure of their pole positions from the trend of Fig. 4. In practice however, tilting of 5° or so might easily be undetectable in the palaeomagnetic pole positions, and this amount of motion would have been more than sufficient to account for all the tensional fracture in the Svecofennian crust.

The formation of the Sveconorwegian Front Zone could have been associated with compressional, transcurrent or vertical motions between the Sveconorwegian and Svecofennian regions, depending on the precise nature of the planar structures which constitute the zone. Neither compressional nor strike-slip motions, however, can account for the simple tensional stresses at right angles to the Front Zone suggested by the 1000–900-m.y. dykes, and if the Front Zone did form by one of these mechanisms, then it was probably formed before 1000 m.y.; in this case, the relative post-orogenic uplift of the Sveconorwegian belt after 1000 m.y. merely took advantage of a pre-existing zone of weakness in the crust. Thus while it is possible that detailed structural and geochronological work could demonstrate a complex pre-1000-m.y. history for the Front Zone, as suggested for the Grenville Front in Canada [5,13], the model relating dyke emplacement to uplift of the Sveconorwegian region applies *only* to times post-dating the main Grenville-Sveconorwegian deformational and metamorphic events, and is therefore independent of the actual age and origin of the Sveconorwegian Front Zone. The Grenville Front has likewise been discussed as a locus of post-orogenic uplift motion not necessarily genetically related to its formation (e.g. [14]).

5. Age of Sveconorwegian and Grenville poles

Table 2 gives the palaeomagnetic Atlantic poles for the dolerites of this study; two dolerites, Karlshamn and Väby, are also covered by published poles, and these are shown in Table 3. The poles from the Sveconorwegian belt and Front Zone given in Table 3 are from either (1) rocks intruded and/or metamorphosed during the Sveconorwegian events, or (2) dolerite dykes known or supposed to have been emplaced during cooling of the province. Relevant ages, mainly

TABLE 3

Published results from southern Scandinavia

No.	Rock unit and area	Tectonic region	Pole position		*dp* (deg.)	*dm* (deg.)	Reference
			lat.	long.			
1a	Karlshamn dolerite ("hyperite")	CR	40°N	27°E	3	3	[40]
4a	Väby dolerite ("hyperite")	CR	8°S	56°E	7	10	[40]
10	Amphibolites and hyperites, Bamble and Kongsberg, Norway	SN	3°N	27°E	(α_{95} = 9)		[40]
11	Amphibolites, norites and anorthosites, Rogaland, Norway	SN	36°N	47°E	10	12	[41]
12	Hunnedalen dolerites, S. Norway	SN	34°N	28°E	9	10	[41]
13	Egersund dolerites, Rogaland, Norway	SN	22°N	51°E	14	16	[41]
14	Farsundite, S. Norway	SN	43°N	14°E	9	10	[42]
15	Tuve dolerite, Gothenburg, S.W. Sweden	SN	17°N	59°E	2	3	[43]
16	Hyperite, S; Sweden	SF	30°N	31°E	2	2	[44]
17	Hyperite, S. Sweden	SF	2°N	33°E	6	8	[44]
18	Hyperite, S. Sweden	SF	27°N	36°E	3	3	[44]
19	Hyperite, S. Sweden	SF	10°N	59°E	2	3	[44]
20	Hyperite, S. Sweden	SF	2°N	61°E	5	8	[44]

Numbers refer to poles plotted in Figs. 4 and 5. Tectonic regions: CR = pre-Sveconorwegian craton, SF = Sveconorwegian Front Zone, SN = Sveconorwegian belt proper.

K–Ar, are listed and referenced by the palaeomagnetic authors given in Table 3.

All poles are plotted in Fig. 4, where it is apparent that the Sveconorwegian poles are essentially co-extensive with the 1000–900-m.y. dolerite poles and define a similar general trend. In detail the southern part of the overall trend is defined by dolerite poles, and the only dolerite result in the northern part of the trend is the Karlshamn dyke where there is both a published and a new pole in close agreement (poles 1 and 1a). If this result were due to remagnetisation in late Sveconorwegian time, then the remaining poles for the dolerites would be co-linear with those from the Sveconorwegian, but the trends would not be superimposed. The demagnetisation curves for the Karlshamn specimens, however, as well as those for the closely neighbouring Tärnö dyke (Fig. 2a and b), suggest that if this had occurred, then the temperature must have been 450°C or more. Reheating to this temperature would be expected to cause alteration of the dolerite and considerable disturbance to the mineral lines: this does not appear to be the case unless complete isotopic resetting of all minerals, including apatite and clinopyroxene has taken place without any alteration or textural recrystallisation. Additionally, the biotite from country rock granite 15 km *closer* to the Front Zone than the Karlshamn dyke shows no evidence of Rb–Sr resetting in Sveconorwegian time, and this does not allow a temperature higher than ca. 300°C, on the basis of the estimated blocking temperature of Alpine biotites [45]. Thus the Karlshamn result must be regarded as an igneous magnetisation unless it is supposed that an original thermal remanence was completely reset at a temperature lower than 300°C, which seems unlikely.

The Karlshamn result and the results for the dolerites in general are therefore taken as representing cooling closely following igneous intrusion, and as the dolerites have ages between 1000 and 900 m.y., then this age can also be assigned to the Sveconorwegian and Front Zone poles, which lie on the same trend (Fig. 4). The poles from the Front Zone "hyperites" could be intrusive magnetisations if the dykes correlated with those in the Svecofennian region (the hyperites of southern Sweden are conspicuously parallel to the dolerites of this study), or alternatively if these dolerites were intruded much earlier, then the poles could be Sveconorwegian remagnetisations imposed at the time of Front Zone deformation, which micas from the Vaggeryd syenite give as probably

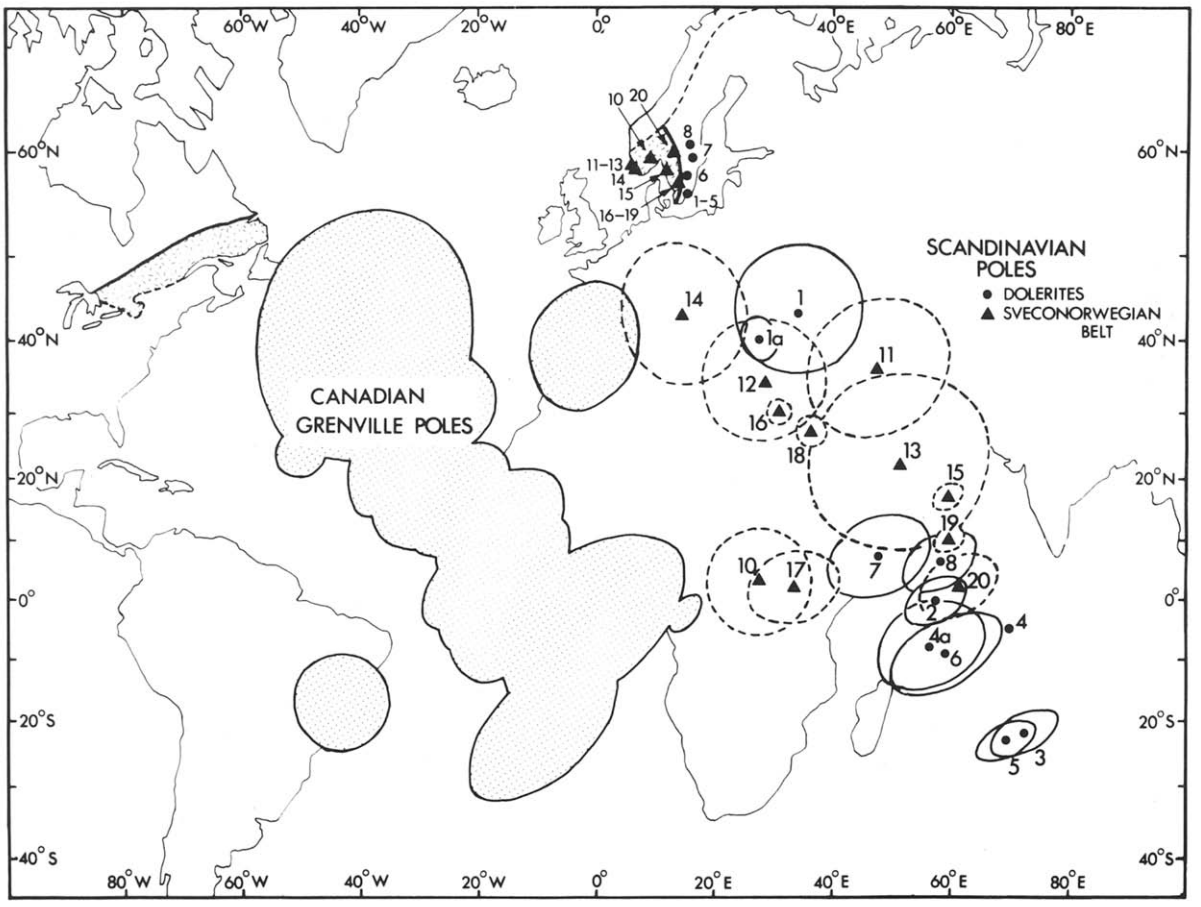


Fig. 4. Mercator projection of the palaeomagnetic poles for the dolerites, Sveconorwegian belt and Front Zone, relative to the area defined by the uncertainties on Grenville poles. The Grenville and Sveconorwegian regions are shown shaded, and the Front Zones as heavy lines.

pre-900 m.y. The question must await a clear demonstration of the age of the Front Zone hyperites (see [46] for some evidence of a pre-1000-m.y. age), but it is clear that they give Sveconorwegian poles, whether of igneous or metamorphic origin.

The 1000–900-m.y. age inferred for the magnetisation in the Sveconorwegian Province, which corresponds to part of the range of K–Ar ages, can also be applied to the Grenville Province. The K–Ar ages for the two regions indicate very similar cooling histories [1,2,10–12], and there is every reason to suppose that magnetisations were acquired at similar times. McWilliams and Dunlop [25] have already tentatively assigned a 1000–900-m.y. age to the Grenville poles, based on K–Ar ages, and the data discussed here support their suggestion.

A 1000–900-m.y. age for magnetisation in the Grenville-Sveconorwegian belt means that the Grenville poles do not correspond in age to any results from pre-Grenville North America, and that they therefore define an additional loop or track in the polar wander path [24,25]. Models which assign pre-1200-m.y. ages to the Grenville poles [19,20], or 1120–1000-m.y. ages [15,17,22,23] are probably no longer applicable. The effect of this is that palaeomagnetic data do not necessitate any plate motion between parts of North America in late Grenville time. However, as the Grenville-Sveconorwegian belt

does not yield any pre-1050-m.y. poles, and since palaeomagnetic data in the interval 1200–1000 m.y. are lacking from Scandinavia, it is not possible at this stage to comment on whether the Grenville was a plate tectonic [47] or an "in situ" type of orogeny involving no significant plate motion [27].

6. North Atlantic reconstruction

In Fig. 4 is shown the general area defined by the uncertainties on Grenville poles, the trend of which is conspicuously similar in direction and length to that of the 1000–900-m.y. Scandinavian results. Scandinavia and its poles were replotted by computer using a clockwise rotation of 66.5° about a pole at 80.5°N, 86.0°W. This rotation pole was identified by an optimum visual fit of the Scandinavian and Canadian palaeomagnetic data using a globe and spherical overlays, coupled with calculation on a stereographic net. The superimposition of the palaeomagnetic poles leads to the continental reconstruction of Fig. 5, essentially the same as that of Ueno et al. [18].

The reconstruction is derived entirely *independently* of the in-detail arrangement of Grenville and Sveconorwegian poles, but the Scandinavian data do lend some tentative support to the "Grenville loop" of polar wander (Fig. 5) postulated by McWilliams and Dunlop [25] on the basis of K–Ar biotite ages. Inspection of Table 1 shows (assuming reliability of ages within errors) that most of the dolerite poles defining the southern part of the trend have ages of 950 m.y. or more. The Karlshamn dyke may be slightly younger than these, giving mineral lines of 956 ± 56 and 890 ± 25 m.y., and this gives poles (numbers 1 and 1a) near the northern limit of the trend. The Tärnö dyke apparently is clearly younger than those from further North in Sweden, giving two mineral lines of ca. 900 m.y., and the pole for this dyke (pole 2) lies in the southern part of the trend. Thus it is possible that there is a loop of polar wander beginning and ending in the southern part of the trend, such that the Karlshamn pole lies close to the hairpin, and the younger end is defined by the Tärnö result (Fig. 5). The ages from the Bräkne-Hoby dyke, which lies close to Karlshamn and Tärnö, have rather high errors, so that it is not possible to assign this pole (pole 3) to any particular part of the loop.

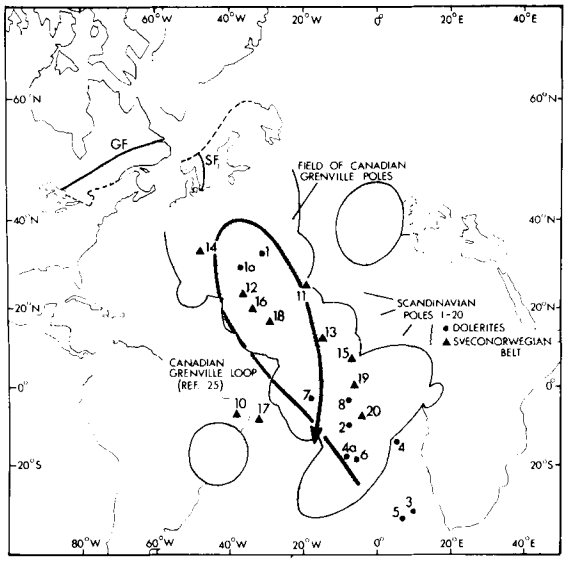


Fig. 5. Mercator projection of the North Atlantic region reconstructed for the time of the close of the Grenville-Sveconorwegian orogeny. GF = Grenville Front, SF = Sveconorwegian Front (Schistosity Zone).

This combined loop of polar wander is highly tentative, subject to the interpretation of differences between the dolerite ages, and it is not possible to assign a direction of motion to it from Scandianavian data. However, the direction of motion suggested from Grenville results [25] is consistent with the Scandinavian poles because it would mean that late or post-tectonic basic dykes from within the Sveconorwegian block (poles 12, 13 and 15, Fig. 5) would fall in the later part of the loop. Any loop fitting the Scandinavian data would differ in detail from the Grenville loop of Fig. 5a: in particular the younger limit of the loop would lie further east to take in poles 2, 13 and 15. Acceptance of this Grenville-Sveconorwegian loop into overall polar wander schemes will depend upon further data and consistency with earlier and later paths.

The loops of polar migration would imply rather rapid drift rates: if the Scandinavian data span 100 m.y., then 15 cm/yr is indicated, while if the maximum span allowed by the age data of 150 m.y. is used, then a value of 10 cm/yr is obtained. This compares to the 8 cm/yr quoted for the proposed Grenville loop [25].

The continental reconstruction of Fig. 5 leaves two possibilities for the geometrical arrangement of the

Grenville-Sveconorwegian belt in the North Atlantic region. First, the Grenville and Sveconorwegian Fronts (GF and SF, Fig. 5), which lie approximately at right angles to each other, could represent the *same* tectonic margin, in which case the belt must have a ca. 90° change in direction between Canada and Scandinavia and be "closed" to the north. Secondly, the two fronts could represent *opposite* tectonic margins, in which case the belt could continue northwards. The fact that no suggestion of the Grenville-Sveconorwegian ages has been found in the central and northern Scandinavian Caledonides tends to favour the model where the fronts are linked and the belt closed to the north. On the other hand, the possible occurrence of Grenville-age rocks in East Greenland [48] may suggest that the belt continues northwards between Greenland and Scandinavia. If it were assumed that northwest Britain lay adjoining southeast Greenland in late Grenville time, then the probable Grenville inliers in Scotland [49] and western Ireland [50] would also support this suggestion. The Grenville-age rocks of such a northward continuation must have all become involved in the later Caledonian events, and in this case a solution to the question must await further detailed geochronology on pre-Caledonian inliers within the Caledonian belt.

The demonstration of a 1000–900-m.y. age for the Grenville-Sveconorwegian poles supports the "supercontinent" reconstruction of Piper [27] for North America and Africa at this time, as the Grenville poles correspond on this hypothesis to 1000–800-m.y. African poles [26]. Together with the America-Scandinavia reconstruction, this implies that Africa, North America and Scandinavia formed a single continental area at the close of the Grenville events. This is reasonable tectonically because "Grenville events" are recorded in southern Scandinavia, eastern Canada, the eastern and southern United States [51], and in central and southern Africa (the Kibaran belt, [52]). The present evidence does not, however, shed light on the question of whether plate tectonics operated in the genesis of these belts, which depends upon palaeomagnetic data between 1300 and 1000 m.y.

Acknowledgements

The authors express thanks to Agnes Rodhe and Zoltan Solyom for kindly making available samples BL-ARX and D75-26 respectively. J. Hutchinson, G. Ross and J. Jocelyn provided skilled technical advice and assistance for isotopic work, and Eva Hanell assisted with the palaeomagnetic measurements. M.R.W. Johnson, O. van Breemen and R. Thompson critically reviewed the manuscript. The financial support of the Natural Environment Research Council (Great Britain) and the Swedish National Research Council for fieldwork and research is acknowledged.

References

1 C.T. Harper, On the interpretation of potassium-argon ages from Precambrian shields and Phanerozoic orogens, Earth Planet. Sci. Lett. 3 (1967) 128.
2 R.M. Macintyre, D. York and W.W. Moorhouse, Potassium-argon age determinations in the Madoc-Bancroft area in the Grenville Province of the Canadian Shield, Can. J. Earth Sci. 4 (1967) 815.
3 H.R. Wynne-Edwards, Tectonic overprinting in the Grenville Province of southwestern Quebec, Geol. Assoc. Can., Spec. Paper 5 (1969) 163.
4 H.R. Wynne-Edwards, The Grenville Province, Geol. Assoc. Can., Spec. Paper 11 (1972) 263.
5 T.E. Krogh and G.L. Davis, The Grenville Front interpreted as an ancient plate boundary, Carnegie Inst. Wash. Yearbook 70 (1971) 239.
6 R.A. Frith and R. Doig, Pre-Kenoran tonalitic gneisses in the Grenville Province, Can. J. Earth Sci. 12 (1975) 844.
7 H.R. Wynne-Edwards and Z. Hasan, Intersecting orogenic belts across the North Atlantic, Am. J. Sci. 268 (1970) 289.
8 N.H. Magnussen, The Precambrian history of Sweden, J. Geol. Soc. Lond. 121 (1965) 1.
9 R.K. O'Nions and H. Baadsgaard, A radiometric study polymetamorphism in the Bamble region, Norway, Contrib. Mineral. Petrol. 34 (1971) 1.
10 N.H. Magnussen, Age determinations of Swedish Precambrian rocks, Geol. Fören. Stockholm Förh. 82 (1960) 407.
11 R.K. O'Nions, R.D. Morton and H. Baadsgaard, Potassium-argon ages from the Bamble sector of the Fennoscandian Shield in south Norway, Nor. Geol. Tiddskr. 49 (1969) 171.
12 T. Skiöld, The interpretation of the Rb–Sr and K–Ar ages of late Precambrian rocks in south-western Sweden, Geol. Fören. Stockholm Förh. 98 (1976) 3.
13 S.J. Brocoum and I.W.D. Dalziel, The Sudbury basin, the Southern Province, the Grenville Front and the Penokean orogeny, Geol. Soc. Am. Bull. 85 (1974) 1571.
14 A.J. Baer, The Grenville Province in Helikian times: a possible model of evolution, Philos. Trans. R. Soc. Lond., Ser. A, 280 (1976) 499.
15 E. Irving and J.C. McGlynn, Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia, Philos. Trans. R. Soc. Lond., Ser. A, 280 (1976) 433.

16 W.F. Fahrig, E. Irving and G.D. Jackson, Paleomagnetism of the Franklin diabase, Can. J. Earth Sci. 8 (1971) 455.
17 E. Irving, R.F. Emslie and H. Ueno, Upper Proterozoic paleomagnetic poles from Laurentia and the history of the Grenville structural province, J. Geophys. Res. 79 (1974) 5491.
18 H. Ueno, E. Irving and R.H. McNutt, Paleomagnetism of the Whitestone anorthosite and diorite, the Grenville polar track, and relative motions of the Laurentian and Baltic Shields, Can. J. Earth Sci. 12 (1975) 209.
19 J.L. Roy and W.F. Fahrig, The paleomagnetism of Seal and Croteau rocks from the Grenville Front, Labrador: polar wandering and tectonic implications, Can. J. Earth Sci. 10 (1973) 1279.
20 W.F. Fahrig, K.W. Christie and E.J. Schwarz, Paleomagnetism of the Mealy Mountain anorthosite suite and of the Shabogamo gabbro, Labrador, Canada, Can. J. Earth Sci. 11 (1974) 18.
21 W.F. Fahrig and D.L. Jones, The paleomagnetism of the Helikian Mistastin pluton, Labrador, Canada, Can. J. Earth Sci. 13 (1976) 832.
22 E. Irving, J.K. Park and J.L. Roy, Paleomagnetism and the origin of the Grenville Front, Nature 236 (1972) 344.
23 K.L. Buchan and D.J. Dunlop, Magnetisation episodes and tectonics of the Grenville Province, Nature Phys. Sci. 246 (1973) 28.
24 A.D. Stewart and E. Irving, Palaeomagnetism of Precambrian sedimentary rocks from NW Scotland and the apparent polar wandering path of Laurentia, Geophys. J. R. Astron. Soc. 37 (1974) 51.
25 M.O. McWilliams and D.J. Dunlop, Precambrian paleomagnetism: magnetisations reset by the Grenville orogeny, Science 190 (1975) 269.
26 J.D.A. Piper, Proterozoic supercontinent: time duration and the Grenville problem, Nature 256 (1975) 519.
27 J.D.A. Piper, Palaeomagnetic evidence for a Proterozoic supercontinent, Philos. Trans. R. Soc. Lond., Ser. A, 280 (1976) 469.
28 N.H. Magnussen, The Swedish Precambrian outside the Caledonian mountain chain, in: N.H. Magnussen et al., Description to accompany the map of the pre-Quaternary rocks of Sweden, Sver. Geol. Unders., Ser. Ba, 16 (1960).
29 G. Bylund and P.J. Patchett, Palaeomagnetic and Rb–Sr isotopic evidence for the age of the Särna alkaline complex, western central Sweden, Lithos 10 (1977) 73.
30 P.J. Patchett and O. van Breemen, unpublished data.
31 P.J. Patchett, Rb–Sr ages by whole-rock and mineral methods of Precambrian dolerites and syenites in Sweden, in preparation.
32 C. Brooks, S.R. Hart and I. Wendt, Realistic use of two-error regression treatments as applied to rubidium-strontium data, Rev. Geophys. Space Phys. 10 (1972) 551.
33 E. Welin and Th. Lundqvist, New Rb–Sr age data for the sub-Jotnian volcanics (Dala Porphyries) in the Los-Hamra region, central Sweden, Geol. Fören. Stockholm Förh. 92 (1970) 35.
34 E. Welin, Den Svecofenniska orogena zonen i norra Sverige – en preliminär diskussion, Geol. Fören. Stockholm Förh. 92 (1970) 433.
35 G. Åberg, An Rb–Sr age for the Småland Porphyries, Geol. Fören. Stockholm Förh. 94 (1972) 311.
36 E. Welin and G. Blomqvist, Further age measurements on radioactive minerals from Sweden, Geol. Fören. Stockholm Förh. 88 (1966) 3.
37 G. Pullaiah, E. Irving, K.L. Buchan and D.J. Dunlop, Magnetisation changes caused by burial and uplift, Earth Planet. Sci. Lett. 28 (1975) 133.
38 R. Gorbatschev, Dolerites of the Eskilstuna region, eastern central Sweden, Sver. Geol. Unders., Ser. C, 580 (1961).
39 S. Hjelmqvist, Beskrivning till berggrundskarta över Kopparbergs län, Sver. Geol. Unders., Ser. Ca, 40 (1966).
40 R.P.E. Poorter, Palaeomagnetism of Precambrian rocks from southeast Norway and south Sweden, Phys. Earth Planet. Inter. 10 (1975) 74.
41 R.P.E. Poorter, Palaeomagnetism of the Rogaland Precambrian (southwestern Norway), Phys. Earth Planet. Inter. 5 (1972) 167.
42 G.S. Murthy and E.R. Deutsch, A new Precambrian palaeomagnetic pole for northern Europe, Phys. Earth Planet. Inter. 11 (1975) 91.
43 N. Abrahamsen, The palaeomagnetic age of the WNW-striking dikes around Gothenburg, Sweden, Geol. Fören. Stockholm Förh. 96 (1974) 163.
44 F.G. Mulder, Palaeomagnetic research in some parts of central and southern Sweden, Sver. Geol. Unders., Ser. C, 653 (1971).
45 E. Jäger, Die alpine Orogenese im Lichte der radiometrischen Alterbestimmung, Eclogae Geol. Helv. 66 (1973) 11.
46 I. Klingspor, Radiometric age-determinations of basalts, dolerites and related syenite in Skåne, southern Sweden, Geol. Fören. Stockholm Förh. 98 (1976) 195.
47 J.F. Dewey and K.C.A. Burke, Tibetan, Variscan and Precambrian basement reactivation: products of continental collision, J. Geol. 81 (1973) 683.
48 B.T. Hansen, F. Oberli and R.H. Steiger, The geochronology of the Scoresby Sund area, central East Greenland. Progress Report 6: Rb/Sr whole-rock and U–Pb zircon ages, Rapp. Grønlands Geol. Unders. 66 (1974) 32.
49 M. Brook, M.S. Brewer and D. Powell, Grenville age for rocks in the Moine of north-western Scotland, Nature 260 (1976) 515.
50 O. van Breemen, D.R. Bowes and W.E.A. Phillips, Evidence for basement of late Precambrian age in the Caledonides of western Ireland, Geology 4 (1976) 499.
51 S.S. Goldich, W.R. Muehlberger, E.G. Lidiak and C.E. Hedge, Geochronology of the midcontinent region, USA, 1. Scope, methods, principles, J. Geophys. Res. 71 (1966) 5375.
52 T.N. Clifford, The structural framework of Africa, in: African Magmatism and Tectonics, T.N. Clifford and I.G. Gass, eds. (Oliver and Boyd, Edinburgh, 1970).